BIO503 MID TERM SOLVED SUBJECTIVE

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Virtual University of Pakistan

BIO503

Biological Physics

Midterm past papers solved

Created By

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Bio503 01-01-2019

Name any two polar compunds? 2 marks

Ans:

- Hydrogen chloride HCL
- Hydrogen Iodide HI

2. Name two metabolic pathways? 2 marks

Ans:

- Catabolic pathway
- Anabolic pathway

3. D/f homonuclear and heteronuclear? 3 marks

Ans: A molecule may be homonuclear, that is, it consists of atoms of one chemical element, as with oxygen (O2) or

It may be heteronuclear, a chemical compound composed of more than one element, as with water (H2O).

4. What is Cardiac electropyisilogy? 3 marks

Ans: Electrophysiology study (EP Test) is used to Study abnormal heart rhythms (arrhythmia).

5. Define diffusion and give some examples also daily life? 5 marks

Ans: The process of diffusion, a substance tends to move from an area of high concentration to an area of low concentration until its concentration becomes equal throughout a space. Examples

- One of the ways substances enter and leave cells is by diffusion
- Diffusion of CO2 into stomata of leaves during photosynthesis.
- Diffusion of O2 into the alveoli of the lungs
- Diffusion of O2 to respiring cells
- Diffusion of glucose, amino acids etc. into the villi during digestion.
- In mammals, birds, reptiles and amphibia, oxygen and carbon dioxide are exchanged by diffusion in the lungs
- In fish, this exchange of gases takes place by diffusion through the gills
- The oxygen dissolved in the water diffuses into the blood vessels in the gills.

6. Write the biophysics of membrane and give importance in biology? 5 marks

Ans: • Membrane biophysics is the study of the physical principles governing biological membranes, including lipid-raft formation and protein–lipid coupling, as well as their mechanical characteristics, and the effect they have on Para cellular transport and phenomena relating to cell shape.

- Membrane biophysics is the study of biological membrane structure and function using physical, computational, mathematical, and biophysical methods.
- Bridge between biology and physics.
- Study of biological systems and biological processes using physics-based methods or based on physical principles.
- Biophysics discovers how atoms are arranged to work in DNA and proteins.
- Biophysicists are discovering how proteins work, they use these protein structures for discovering how biological machines work, in health and also in diseases.

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1) **What is enzyme? 2**

Ans: Enzymes act as catalysts in biochemical reactions, meaning that they speed the reactions up.

2) **what is viscosity? 2**

Ans: Viscosity is the measure of resistance of the fluid to flow and it differs from one fluid to other.

3) **3 types of spectroscopy? 3**

Ans:
- Atomic Emission spectroscopy
- Atomic Absorption spectroscopy
- Scattering spectroscopy

4) **ADP+Pi complete the reaction? 3**

Ans: ADP + P i →ATP + H 2 O

G° = 29 kJ/mol (Endergonic reaction)

5) **what is covalent bond? Explain. 5**

Ans: A covalent bond, also called a molecular bond, is a chemical bond that involves the sharing of electron pairs between atoms

6) **how substance move in and out the cell? 5**

Ans: • Osmosis is a type of diffusion that involves the random movement of water molecules.

- It is the movement of water across a partially-permeable membrane, from an area of high water concentration to an area of low water concentration.
- Osmosis explains how water moves in and out of animal cells through the cell membrane.
- If water continues to leave a cell it will shrink and become crinkly. If water continues to enter a cell it will eventually burst.
- These things happen because animal cells are not surrounded by an inelastic cell wall, unlike plant cells.
- Animal cells do not have cell walls.
- In hypotonic solutions, animal cells swell up and explode as they cannot become turgid because there is no cell wall to prevent the cell from bursting.
- When the cell is in danger of bursting, organelles called contractile vacuoles will pump water out of the cell to prevent this.
- In hypertonic solutions, water diffuses out of the cell due to osmosis and the cell shrinks. Thus, the animal cell has always to be surrounded by an isotonic solution.
- In the human body, the kidneys provide the necessary regulatory mechanism for the blood plasma and the concentration of water and salt removed from the blood by the kidneys is controlled by a part of the brain called the hypothalamus.
- The process of regulating the concentration of water and mineral salts in the blood is called osmoregulation.
- Animals which live on dry land must conserve water and so must animals which live in the salty sea water, but animals which live in freshwater have the opposite problem; they must get rid of excess water as fast as it gets into their bodies by osmosis.

bio503 24-12-2018

Subjective part

1-wirte a note on sliding friction. example?5

Ans: Sliding friction occurs between objects as the slide against each other. It acts in the direction opposite to the direction of motion. It prevents the object from moving too fast. It is also called as kinetic friction.

- Examples: A book sliding from an inclined desk, A kid sliding on a slide.

- When sliding friction is acting there must be another force existing to keep the body in moving, in case of the book sliding from the desk the other force acting is gravitational force.

2-explain fahrenheight, celsius and kelvin scale?5

Ans:
Fahrenheit: This is one of the oldest scales, and is based on the coldest temperature that could be achieved with a mix of ice and alcohol. In it the freezing point of water is at 32 °F, the boiling point of water is at 212 °F.

- **Celsius or Centigrade:** This is a very sane system, where the freezing point of water is at 0 °C and the boiling point is at 100 °C. The degree size is thus as big as the Fahrenheit degree.
- **Kelvin or Absolute:** 0 K is the lowest possible temperature, where the internal energy of a system is at its absolute minimum. The degree size is the same as that of the Centigrade or Celsius scale. This makes the freezing point of water at atmospheric pressure 273.16 K, the boiling point at 373.16 K

3-Mention three part of mass spectrometer?3

Ans: Mass spectrometers can be divided into three fundamental parts, namely

- Ionization source
- Analyzer
- Detector

4-why does a hydrogen bonding occur?3

Ans: Since the hydrogen donor is strongly electronegative, it pulls the covalently bonded electron pair closer to its nucleus, and away from the hydrogen atom. The hydrogen atom is then left with a partial positive charge, creating a dipole-dipole attraction between the hydrogen atom bonded to the donor, and the lone electron pair on the acceptor. This results in a hydrogen bond.

5-name two types of hydrogen bond?2

Ans:

1. Intramolecular hydrogen bonds
2. Intermolecular hydrogen bonds

6-define rotatory motion?2

Ans: Rotatory motion is anything that moves in a circle.

Objective part

1-which one is not used as solvent in MNR?

2-the average bond-energy for O-H bond in water is..... 463kJ/mol.

3-..... covers all scales of biological organization, from molecular to organismic and population.

4-biophysicists discover the DNA double helix in 1953

5-electrophysiology involves the movement of voltage and current through the ..... ion channel protein

6-nitrogen gas is diatomic and nonpolar because both the nitrogen atoms have .....same degree of electronegativity.

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what are enzymes?

Ans: Enzymes are very efficient catalysts for biochemical reactions. They speed up reactions by providing an alternative reaction pathway of lower activation energy.

define the 2nd law of thermodynamics.

Ans: Every energy transfer or transformation increases the entropy of the universe.

Explain the hydrodynamics

Ans: The scientific study of the motion of fluids, especially non-compressible liquids, under the influence of internal and external forces.

- The behavior of large biomolecules—proteins, carbohydrates, and nucleic acid in solution is complex and directly related to molecular size, shape, and flexibility.
- Analysis of hydrodynamic behavior thus provides important information about the structure, dynamics, and interactions of bio macromolecules.

Random motion of moving faster molecule RNA function? or characteristics ko channel protein?

Ans: Integral lipoproteins which contain a pore via which ions may cross from one side of the membrane to the other

- Channel proteins are ion-selective and may be gated to regulate the passage of ions in response to certain stimuli • Channel proteins only move molecules along a concentration gradient (i.e. are not used in active transport)
- Channel proteins have a much faster rate of transport than carrier proteins

bio503 8:30am

state equation of Enthalpy? 2 marks

Ans: The enthalpy of a system is equal to the system's internal energy plus the product of its pressure and volume.

$H = E + PV$

Name two main methods of Frictationations? 2 marks

Ans:
1.Bioassay-guided fractionation

2.Blood fractionation

Second law of thermodynamics in term of entropy? 3 marks

Ans: "The Second Law of Thermodynamics states that when energy is transferred, there will be less energy available at the end of the transfer process than at the beginning"

- Due to entropy, which is the measure of disorder in a closed system, all of the available energy will not be useful to the organism. Entropy increases as energy is transferred.

Anabolic pathways with examples? 3 marks

Ans: Anabolic pathways consume energy to build complex molecules from simpler ones. The synthesis of protein from amino acids is an example of anabolism

Writes types of RNA? 5 marks

Ans: There are several different types of RNA.

- **Messenger RNA (mRNA)** is the RNA transcript or RNA copy of the DNA message produced during DNA transcription. Messenger RNA is translated to form proteins.
- **Transfer RNA (tRNA)** has a three dimensional shape and is necessary for the translation of mRNA in protein synthesis.
- **Ribosomal RNA (rRNA)** is a component of ribosomes and is also involved in protein synthesis.

MicroRNAs (miRNAs) are small RNAs that help to regulate gene expression.

Write above spin spin coupling? 5 marks

Ans: The effective magnetic field is also affected by the orientation of neighboring nuclei. This effect is known as spin-spin coupling which can cause splitting of the signal for each type of nucleus into two or more lines. The size of the splitting (coupling constant or J) is independent of the magnetic field and is therefore measured as an absolute frequency (usually Hertz). The number of splitting indicates the number of chemically bonded nuclei in the vicinity of the observed nucleus.

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Define photosynthesis (2)

Ans: In photosynthesis, the energy is supplied by the sun. Light energy is absorbed by cells in plant leaves and converted to chemical energy.

Who is Karl Pearson (2)

Ans: An English mathematician and biostatistician.

Law of thermodynamics (3)

Ans: Thermodynamics is the field of physics that deals with the relationship between heat and other properties (such as pressure, density, temperature, etc.) in a substance.

Name the instrumentation of HPLC (5)

Ans: HPLC instrumentation includes a pump, injector, column, detector and integrator or acquisition and display system. The heart of the system is the column where separation occurs.

Components:

- Solvent Reservoir
- Pump
- Sample Injector
- Columns
- Detector
- Data Collection Devices

- **Solvent Reservoir :** Mobile phase contents are contained in a glass reservoir. The mobile phase, or solvent, in HPLC is usually a mixture of polar and non-polar liquid components whose respective concentrations are varied depending on the composition of the sample.
- **Pump :** A pump extracts the mobile phase from the solvent reservoir and forces it through the system's column and detector. Depending on a number of factors including column dimensions, particle size of the stationary phase, the flow rate and composition of the mobile phase, operating pressures of up to 42000 kPa (about 6000 psi) can be generated.
- **Sample Injector :** The injector can be a single injection or an automated injection system. An injector for an HPLC system should provide injection of the liquid sample within the range of 0.1-100 mL of volume with high reproducibility and under high pressure (up to 4000 psi).
- **Columns :** Columns are usually made of polished stainless steel, are between 50 and 300 mm long and have an internal diameter of between 2 and 5 mm. They are commonly filled with a stationary phase with a particle size of 3–10 µm. Columns with internal diameters of less than 2 mm are often referred to as microbore columns. Ideally the temperature of the mobile phase and the column should be kept constant during an analysis.
- **Detector :** The HPLC detector, located at the end of the column detect the analytes as they elute from the chromatographic column. Commonly used detectors are UVspectroscopy, fluorescence, mass-spectrometric and electrochemical detectors.
- **Data Collection Devices :** Signals from the detector may be collected on chart recorders or electronic integrators that vary in complexity and in their ability to process, store and reprocess chromatographic data. The computer integrates the response of the detector to each component and places it into a chromatograph that is easy to read and interpret.

Bio503, 24 December 2018, 12:00 pm

1) **State function of RNA. (2 marks)**

Ans: Ribonucleic acid (RNA) is a polymeric molecule essential in various biological roles in coding, decoding, regulation, and expression of genes.

2) **Define ion exchange chromatography. (2 marks)**

Ans: Ion exchange chromatography involves the separation of ionizable molecules based on their total charge.

3) **Why does hydrogen bonding occur? (3 marks)**

Ans: Answer above

4) **Catabolic pathways. (3 marks)**

Ans: Catabolic pathways release energy by breaking down complex molecules into simpler compounds

- Cellular respiration, the breakdown of glucose in the presence of oxygen, is an example of a pathway of catabolism

5) **Open system. (5 marks)**

Ans: An open system is a system that freely exchanges energy and matter with its surroundings.

- For instance, when you are boiling soup in an open saucepan on a stove, energy and matter are being transferred to the surroundings through steam.
- The saucepan is an open system because it allows for the transfer of matter (for example adding spices in the saucepan) and for the transfer of energy (for example heating the saucepan and allowing steam to leave the saucepan).

6) **Write down the principal of gel filtration chromatography. (5 marks)**

Ans: • SEC works by trapping smaller molecules in the pores of the adsorbent materials adsorption ("stationary phases").

- This process is usually performed with a column, which consists of a hollow tube tightly packed with extremely small porous polymer beads designed to have pores of different sizes. These pores may be depressions on the surface or channels through the bead.
- As the solution travels down the column some particles enter into the pores. Larger particles cannot enter into as many pores.
- The larger the particles, the faster the elution. The larger molecules simply pass by the pores because those molecules are too large to enter the pores.
- Larger molecules therefore flow through the column more quickly than smaller molecules, that is, the smaller the molecule, the longer the retention time.

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Ionic bond ?2

Ans: Ionic bonding is a type of chemical bond that involves the electrostatic attraction between oppositely charged ions.

Three function of RNA?3

Ans: Answer above

Hypotonic, Hypertonic , isotonic?3

Ans:

If the water concentration of the cells cytoplasm is lower then that of the medium (i.e.

the medium is a **hypotonic solution**
If the water concentration inside the cell is higher then that of the medium is a **hypertonic solution**

If the water concentration inside the cell is the same as that in the surrounding medium is a **isotonic solution**

NMR instruments details ?5

Ans:
1.Sample holder :- Glass tube with 8.5 cm long,0.3 cm width

2.Permanent magnet :- It provides homogeneous magnetic field at 60-100 MHZ **3.Magnetic coils :-** These coils induce magnetic field when current flows through them. **4.Sweep generator :-** To produce the equal amount of magnetic field pass through the sample

5.Radio frequency (radio transmitter coil) :- A radio transmitter coil that produces a short powerful pulse of radio waves

6.Radio frequency (radio receiver coil):- A radio receiver coil that detects Receiver radio frequencies emitted as nuclei relax to a lower energy level **7.Readout system :-** A computer that analyses and record the data

Three Inthalpy unit?

Ans: The unit of measurement for enthalpy in the International System of Units (SI) is the joule.

• Other historical conventional units still in use include the British thermal unit (BTU) and the calorie.

Name of main energy carriers by cell?

Ans: The main chemical energy carrier of the cell is ATP, short for adenosine triphosphate, and cells convert energy from ATP to many things.

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